Revisiting the Platonic Representation Hypothesis

Results to date: E1 simulation, E2 audit of 20 LLMs, E3 Rosetta yardstick, E6 global representations · 12 October 2026

Summary

1. Their calibration threshold measures dimensionality. The Aristotelian calibrated test takes a maximum over all layer pairs. Its threshold equals the chance level of the highest-effective-rank layer pair ($r = 0.99999$ across 20 LLMs × 17 vision models), and it rises with model size inside every family. So their calibrated scaling trend partly measures dimensionality, not convergence.
2. Calibrating a single fixed pair is harmless. On Rosetta's benchmark (420 unpaired alignments), calibrated, raw and unbiased CKA predict real alignment equally well (pre-registered H1: null). Rosetta's own result replicates almost exactly ($\rho = -0.895$, $R^2 = 0.75$).
3. Generative decoders hide their global structure. Unpaired alignment error is 0.40 for the generative decoder vs 0.21 for its embedding-model counterpart. Clipping massive activations masks this: the decoder's CKA rises from 0.35 to 0.49.
4. A prompted global readout brings back convergence with scale. On Qwen3 (0.6B → 8B), a training-free PromptEOL readout gives calibrated CKA 0.23 → 0.37 under the Aristotelian pipeline itself. Mean pooling (the readout Huh et al. and the Aristotelian paper use) stays flat at 0.25–0.27. This is so far one model family; replicating it across 20 LLMs is the next experiment.

Experiments at a glance

	Question	Scale	Status
E1	Does the calibrated test recover a known planted alignment?	Simulation, n = 1,024, widths 256–8,192	Done
E2	Replication and mechanism on real models; trends with scale	20 LLMs × 17 ViTs, WIT-1,024, 200 permutations, exact bf16	Done
E3	Which similarity score predicts real unpaired alignability?	84 combinations × 5 seeds = 420 Rosetta alignments	Done
E6	Does a global language representation change the picture?	7 language models × 3 readouts × 6 ViTs × 2 readouts	Done
E2-prompt	Does the prompted readout restore convergence across 20 LLMs?	Re-extraction with PromptEOL	Next

E1 · Simulation: the calibrated score depends on effective rank

We planted a shared latent signal with a known population CKA of 0.30 and varied the feature width, using the Aristotelian recipe (n = 1,024, 200 permutations, α = 0.05). With no shared signal, every calibrated score is about 0, so the calibration does its stated job. But with high-effective-rank features, the calibrated score falls as width grows, while the truth stays fixed.

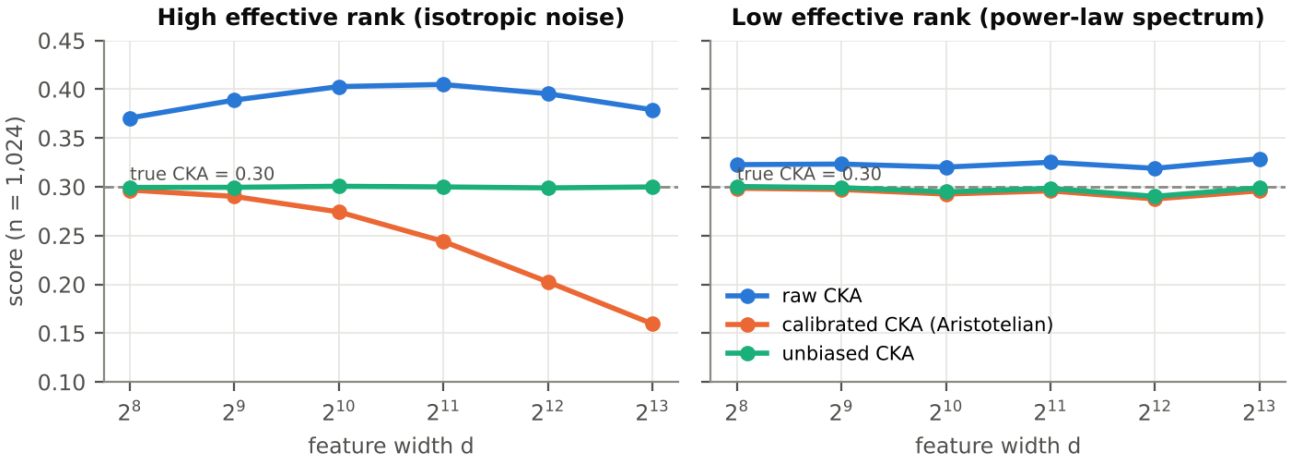


Figure 1. Calibrated CKA (orange) underestimates a fixed true alignment of 0.30 more and more as width grows when features have high effective rank (left), falling from 0.30 to 0.16. With a low-rank, power-law spectrum (right) there is no penalty. Unbiased CKA (green) matches the truth in every setting. Means over replicates.

Closed form (validated against the simulation, r = 0.954, mean error 0.005): raw CKA ≈ (CKA_{true} + $\sqrt{(r_x r_y)}/n$) / $\sqrt{((1 + r_x/n)(1 + r_y/n))}$, where r is the effective rank. Calibration removes the additive chance term but not the shrinkage, which matters once r/n ≳ 0.5.

Other E1 results:

- Depth penalty (more layers to search) is not supported: with correlated layers the max-over-layers threshold does not grow from 8 to 80 layers.
- Mutual k-NN is not more powerful than CKA at detecting a shared signal.
- A few high-variance dimensions with no shared signal sharply cut global alignment (0.30 → 0.02); local mutual k-NN degrades less.
- Small-bandwidth RBF-CKA is degenerate: raw ≈ 1 and calibrated ≈ 0 even with a planted 0.30 alignment.

E2 · Real-model audit: 20 LLMs × 17 vision models

We used the Huh et al. / Aristotelian pipeline (WIT-1,024; 95th-percentile clipping plus L2 normalisation; maximum over all layer pairs; permutation null on the maximum; 200 permutations), confirmed against the official Aristotelian code. All 20 LLMs ran in exact bf16; the largest models used CPU offload, which is bit-identical to an all-GPU run. LLaMA-65B in 4-bit and in bf16 agree within 0.002 on every metric.

Their headline reproduces

Size-rank correlation (original 12 LLMs)	Ours	Aristotelian paper
Raw CKA	0.87	0.86
Calibrated CKA	0.47	0.45
Mutual k-NN	0.85	0.85

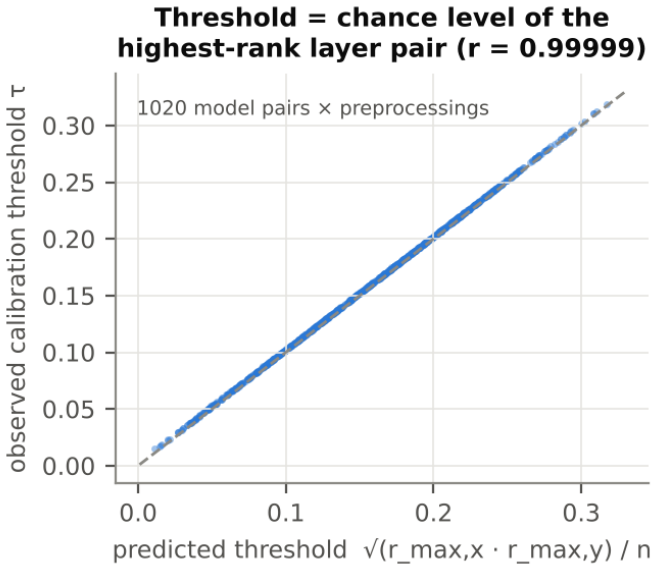


Figure 2. The calibration threshold τ is predicted exactly by the chance level of the highest-effective-rank layer pair on each side (r = 0.99999, mean error 0.002, all model pairs and preprocessings). It is set by dimensionality alone.

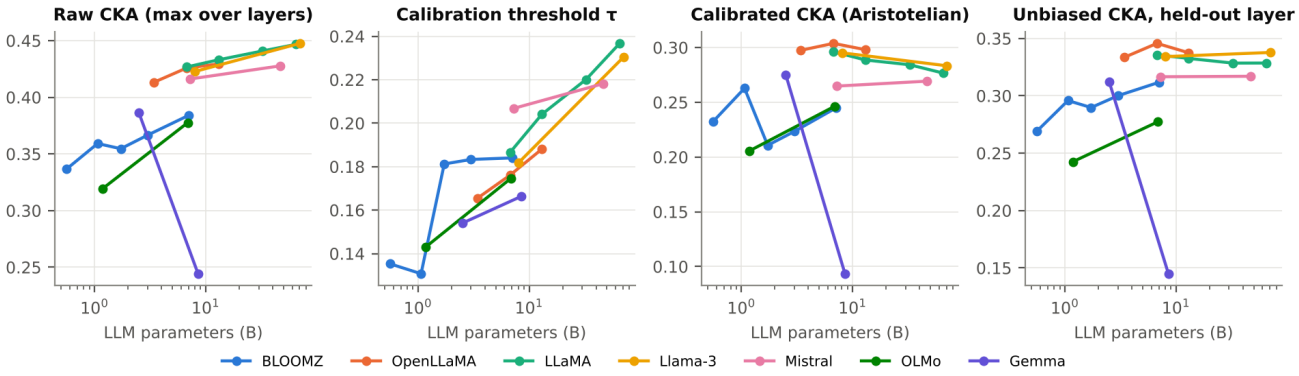


Figure 3. Per family (mean over 17 vision models, Huh preprocessing). The threshold τ rises with size in every family (second panel), so raw CKA rises and calibrated CKA falls in step with it, e.g. Llama-3 8B → 70B: τ 0.18 → 0.23, calibrated CKA 0.30 → 0.28. Held-out unbiased CKA is nearly flat at the top (Llama-3 0.334 → 0.338). Gemma-7B is an outlier dominated by a massive-activation dimension.

Statistics with a two-way bootstrap (resampling LLMs and vision models)

Claim	Evidence	Verdict
Threshold tracks effective rank and rises with size	r = 0.99999; rises in all 7 families	Strong
Local alignment (mutual k-NN) rises with capability	r = 0.73 [0.40, 0.97]	Solid

Claim	Evidence	Verdict
Calibrated / unbiased CKA rise with capability	0.37 [0.14, 0.78] / 0.30 [0.04, 0.90]; difference −0.07 [−0.28, +0.12]	Weak; no difference between estimators
Saturation (kink at 7B)	Above − below slope CI includes 0 for every metric	Not established (descriptive only)
Earlier CIs (vision models only)	About 10× too narrow	Corrected

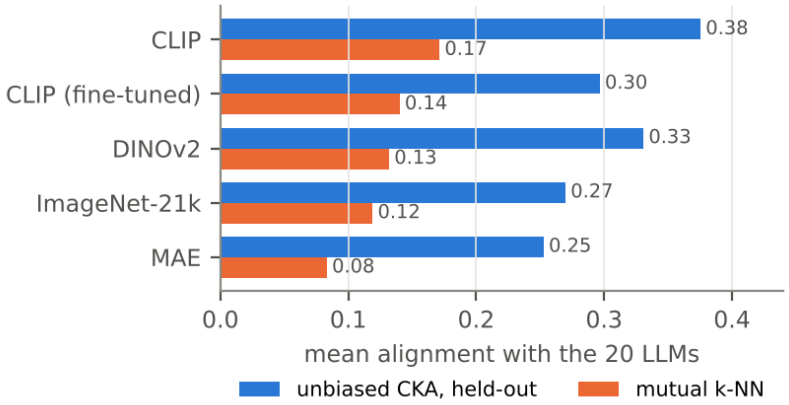


Figure 4. Vision-side objective matters: MAE (masked reconstruction) aligns worst with the LLMs, DINOv2 (global self-distillation) better, and CLIP best (it was trained with language, so partly by construction).

E3 · Rosetta yardstick: which score predicts real unpaired alignment?

Outcome: FOSCTTM of Rosetta's unpaired Wasserstein-Procrustes aligner (their code, unchanged) on all 40,504 MS COCO validation pairs, averaged over 5 seeds. Combinations: 4 training corpora × 7 vision models × 3 language models (mpnet, Qwen3-Embedding-8B, Qwen3-8B generative). Aristotelian measurements used their code at n = 1,024, K = 500, averaged over 20 random subsets. All 420 alignments completed.

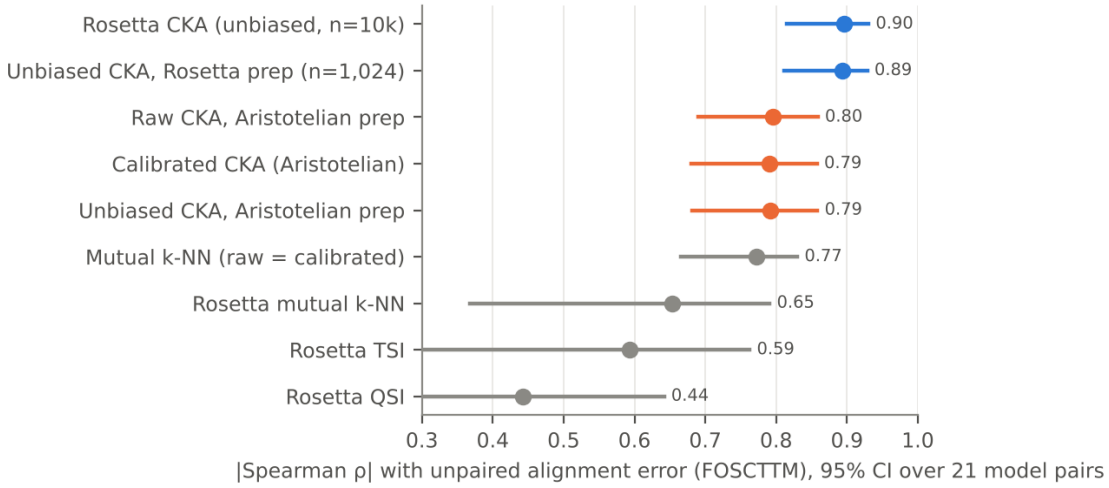


Figure 5. Predictive validity of each score (|Spearman ρ| with mean FOSCTTM over 84 combinations; 95% cluster-bootstrap CI over the 21 model pairs). Blue: unbiased CKA with Rosetta preprocessing; orange: the Aristotelian pipeline (clipping); grey: neighbourhood and topology scores.

Pre-registered test	Δ|ρ| [95% CI]	Verdict
H0: replicate Rosetta (paper R^2 0.73, ρ −0.90)	R^2 0.752, ρ −0.895	Replicated
H1: unbiased − calibrated CKA (same preprocessing)	+0.002 [−0.002, +0.006]	Null: calibrating a fixed pair is harmless
H1b: raw − calibrated CKA	+0.006 [−0.007, +0.023]	Null
H2: calibration does not change mutual k-NN	−0.000 [−0.002, 0.000]	Confirmed
S1: n = 10,000 vs n = 1,024	+0.002 [−0.001, +0.007]	Sample size irrelevant
Exploratory: Rosetta − Aristotelian preprocessing	+0.102 [+0.042, +0.170]	Clipping costs ~0.10 (see below)

The generative decoder aligns about twice as badly, and clipping hides it

Language model	Mean FOSCTTM	CKA (Rosetta prep)	CKA (clipped)	Calibrated CKA
mpnet (encoder)	0.212	0.541	0.524	0.521
Qwen3-Embedding-8B	0.206	0.583	0.567	0.565
Qwen3-8B generative (decoder)	0.399	0.347	0.494	0.490

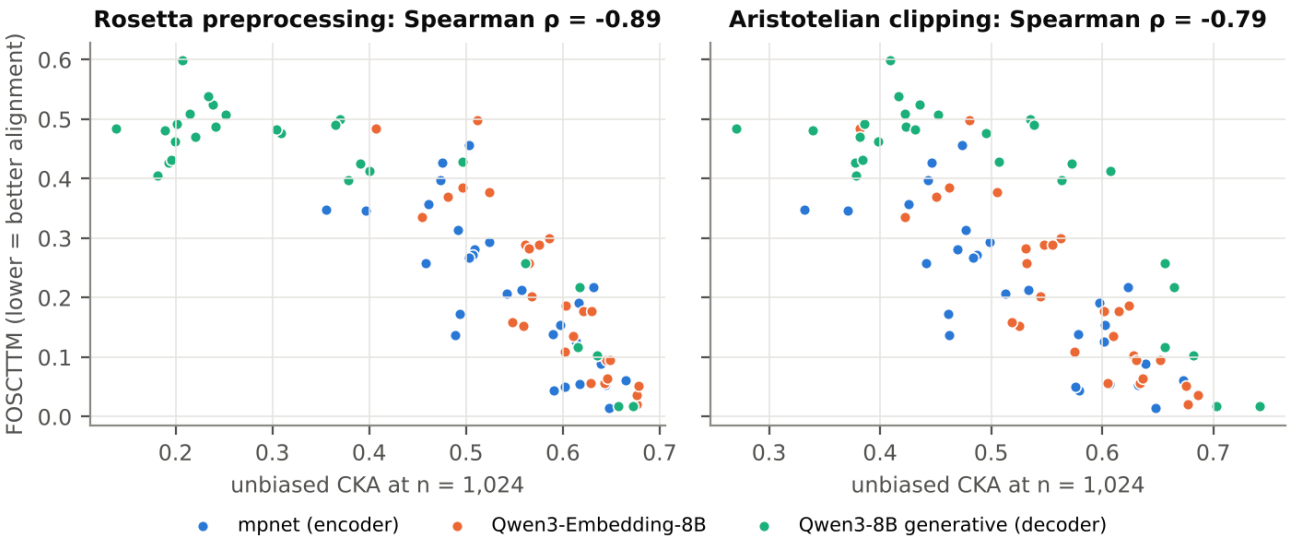


Figure 6. Each point is one combination. With Rosetta preprocessing (left) the decoder's poor alignment shows up as low CKA. Clipping (right) lifts the decoder's CKA to the encoders' level while its alignment stays poor, which accounts for the whole 0.10 loss in predictive power. Within the encoders, every CKA variant predicts equally well ($\rho \approx -0.82$). The decoder is worse in 23–24 of 28 matched comparisons ($p < 1e\text{-}5$); COCO is the one corpus where it is comparable.

E6 · Global representations: decoders vs encoders, MAE vs DINOv2

We tested Mayug's hypothesis (global structure survives calibration if the language side is a global representation) with the E2 engine on WIT-1,024. Language: Qwen3-Base decoders at 0.6B / 4B / 8B, read out by mean pooling, last token, or a PromptEOL prompt (the last token of *This sentence :* "..." *means in one word:*"); Qwen3-Embedding at the same sizes; mpnet. Vision: MAE-B/L/H and DINOv2-B/L/g, each read by [CLS] and by patch mean. The checker passed 16/16.

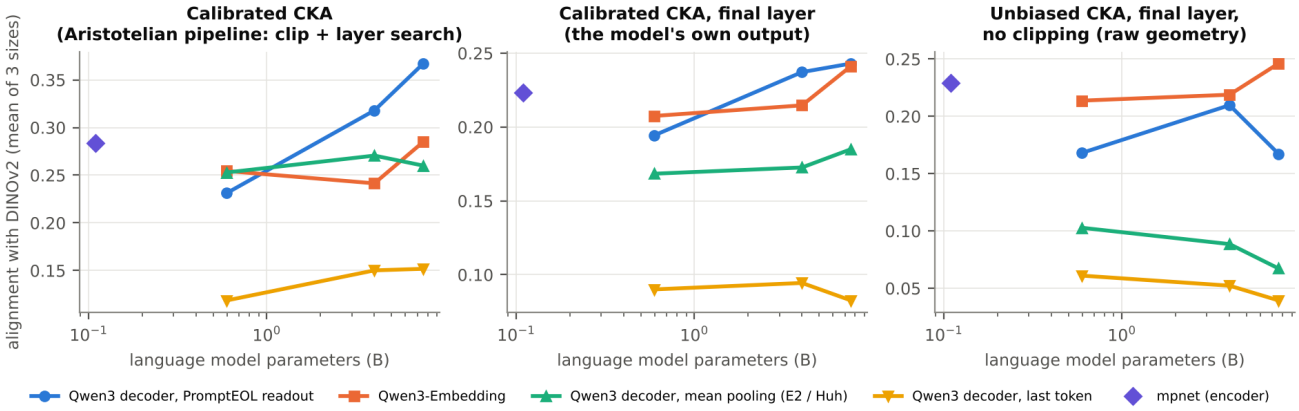


Figure 7. Alignment with DINOv2 ([CLS], mean of 3 sizes) by language readout and size. Left: under the Aristotelian pipeline, the PromptEOL readout rises steeply with size (calibrated CKA 0.23 → 0.32 → 0.37), while mean pooling and the embedding models stay nearly flat. Middle: at the final layer, embedding models beat mean-pooled decoders in 9/9 comparisons. Right: in the raw, unclipped geometry the gap is large and mean-pooled decoders get worse with size.

Pre-registered prediction	Result	Verdict
P1 (primary): embedding > decoder-mean, layer search + clipping	calibrated −0.001 [−0.010, +0.006]; held-out unbiased +0.000	Not supported
P1 (secondary): same, final layer / no clipping	+0.046 (9/9) / +0.14 to +0.18 (9/9)	Supported
P2: embedding models converge faster with scale	slopes equal (+0.011 vs +0.011 per doubling)	Not supported
P3: encoder > decoder also with MAE (vs "like with like")	+0.007 (6/9) clipped; +0.14 (9/9) unclipped	Supported; "like with like" rejected
P4: prompt / last-token readout ≥ mean	prompt +0.05 (9/9); bare last token −0.09 (0/9)	Prompt supported; last token is worst
Exploratory: PromptEOL slope with scale	+0.036 per doubling [0.031, 0.039] vs +0.003 for mean pooling	Strong, one family

Reading: decoders do carry global structure in their intermediate layers, about as much as embedding models once massive activations are clipped. The deficit is in the readout (mean pooling, last token, final layer) and in the raw geometry. Asking the decoder for a global summary through a prompt exposes the structure, and it is the only readout under which alignment grows clearly with scale.

What this adds up to

Claim	Evidence	Strength
Calibrated PRH trend confounds alignment with dimensionality	E1 closed form; E2 τ = chance level (r = 0.99999), rising in every family	Very strong
Calibrating a fixed pair is harmless	E3 H1 null (+0.002 [−0.002, +0.006])	Strong (clean null)
Decoders hide global structure in their readout and raw geometry	E3 decoder vs embedding (0.40 vs 0.21); E6 final layer and unclipped	Strong, one model family for the paired comparison
Clipping masks the decoder deficit	E3 (+0.10 Δ|ρ|, entirely from the decoder); E6 unclipped gaps 3× larger	Strong
A global (prompted) readout restores convergence with scale	E6 PromptEOL: 0.23 → 0.37 calibrated	Promising, one family
Global convergence saturates at the top	E2 plateau, not significant under two-way bootstrap	Descriptive only

Next steps

- **E2-prompt (decisive):** re-extract all 20 E2 LLMs with the PromptEOL readout and rerun the audit with the two-way bootstrap. Include prompt controls: a second summarising prompt, a neutral prompt of the same length, and the caption repeated twice. Add a second dataset (COCO or DCI captions).
- **Rosetta–PromptEOL:** unpaired alignment with prompted decoder embeddings. Does a training-free readout close the 0.40 → 0.21 FOSCTTM gap?
- **Mechanism:** effective rank and massive-activation share of the prompted readout; a learned text-only global embedding as a comparison to the prompt.
- **Mayug meeting (Wednesday):** compare setups (readouts, clipping, layer search) and agree framing and credit.
- **CVPR 2027:** abstract 10 Nov, paper 16 Nov. Go/no-go on the headline after E2-prompt and Rosetta–PromptEOL, around 25 Oct.

Caveats

- WIT-1,024 is a single dataset of 1,024 pairs; E3 uses Rosetta's corpora, but E2 and E6 do not yet have a second dataset.
- The E6 readout results come from one LLM family (Qwen3) at three sizes, with 3 vision models per family, so confidence intervals are rough.
- With 20 LLMs, the two-way bootstrap leaves trend and saturation claims wide; only the mechanism and the mutual k-NN trend are tight.